

MLDM Assignment 3

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MLDM Assignment 3

This are your test results

 Recall

☒ Identify matrix properties

Status

Seen but not answered

Your score

0 / 2

0%

Response

From the below statements, select all the statements that belong to an identity matrix.

Note:

Correct answer: 1 point

Incorrect answer: -0.5 point

Unanswered: 0 points

Max score: 2 points

Min score: 0 points

☐ The product of a matrix with its inverse is always an identity matrix.

☐ All identity matrices are orthogonal matrices.

☐ Every identity matrix has its principal diagonal elements as zeros.

☐ All idempotent matrices are identity matrices.

 Unit vector properties

Status

Not started

Your score

0 / 4

0%

Response

From the statements below, choose whether each statement is 'Right' or 'Wrong'.

Note:

Correct answer: 1 point

Incorrect answer: -0.5 points

Unanswered: 0 points

Max score: 4 points

Min score: 0 points

Unanswered	Right	Wrong	
☐	☐	☐	The norm of the unit vector is always 1.
☐	☐	☐	The vector sum of two unit vectors is always a unit vector.
☐	☐	☐	The cross-product of two orthogonal unit vectors is a unit vector.
☐	☐	☐	The dot product of two unit vectors is equal to the sine of the angle between them.

https://olat.vcrp.de/auth/RepositoryEntry/3909322134/CourseNode/106528575054210/test/0

1/8

☰ Determinant properties			
Status	Not started		
Your score	0 / 3		0%

Response

Among the following statements, select which are 'right' and which are 'wrong'?

Note:

- Correct answer: 1 point
- Incorrect answer: -0.5 points
- Unanswered: 0 points
- Max score: 3 points
- Min score: 0 points

Unanswered	Right	Wrong	
☐	☐	☐	The determinant can only be defined for square matrices.
☐	☐	☐	For any two matrices A and B , $\det(A + B) = \det(A) + \det(B)$
☐	☐	☐	The determinant of the identity matrix is always 1.

📊 3x3 Det			
Status	Not started		
Your score	0 / 2		0%

Response

Compute the determinant of the below matrix A and provide your value in the box given.

Note:

- Correct answer: 2 points
- Incorrect answer: 0 points
- Unanswered: 0 points
- Max score: 2 points
- Min score: 0 points

$$B = \begin{bmatrix} 19 & 56 & 78 \\ 60 & 50 & 22 \\ 15 & 20 & 1 \end{bmatrix}$$

$|B| =$

📊 2x2 Det			
Status	Not started		
Your score	0 / 1		0%

Response

Compute the determinant of the below matrix A and provide your answer in the box given.

Note:

- Correct answer: 1 point
- Incorrect answer: 0 points
- Unanswered: 0 points
- Max score: 1 point
- Min score: 0 points

$$A = \begin{bmatrix} 20 & 10 \\ 4 & 20 \end{bmatrix}$$

$|A| =$

☰ Probablility basic properties																							
Status	Not started																						
Your score	0 / 4		0%																				
Response Select whether the below statement is 'right' or 'wrong'. Note: • Correct answer: 1 point • Incorrect answer: -0.5 points • Unanswered: 0 points • Max score: 4 points • Min score: 0 points <table border="1"> <thead> <tr> <th>Unanswered</th> <th>Right</th> <th>Wrong</th> <th></th> </tr> </thead> <tbody> <tr> <td>☐</td> <td>☐</td> <td>☐</td> <td>The joint probability of two mutually exclusive events is always zero.</td> </tr> <tr> <td>☐</td> <td>☐</td> <td>☐</td> <td>$P(A B) = P(A \cap B)/P(A)$</td> </tr> <tr> <td>☐</td> <td>☐</td> <td>☐</td> <td>The sum of the probabilities of two complementary events is always equal to one.</td> </tr> <tr> <td>☐</td> <td>☐</td> <td>☐</td> <td>The joint probability of two independent events is always zero.</td> </tr> </tbody> </table>				Unanswered	Right	Wrong		☐	☐	☐	The joint probability of two mutually exclusive events is always zero.	☐	☐	☐	$P(A B) = P(A \cap B)/P(A)$	☐	☐	☐	The sum of the probabilities of two complementary events is always equal to one.	☐	☐	☐	The joint probability of two independent events is always zero.
Unanswered	Right	Wrong																					
☐	☐	☐	The joint probability of two mutually exclusive events is always zero.																				
☐	☐	☐	$P(A B) = P(A \cap B)/P(A)$																				
☐	☐	☐	The sum of the probabilities of two complementary events is always equal to one.																				
☐	☐	☐	The joint probability of two independent events is always zero.																				

🎲 Fair dice			
Status	Not started		
Your score	0 / 1		0%
Response A fair dice is rolled two times consecutively. What is the probability of getting one six and one three in a row? Provide your answer in the given box. Please round your answer to three digits after the decimal point. Note: • Correct answer: 1 point • Incorrect answer: 0 points • Unanswered: 0 points • Max score: 1 point • Min score: 0 points Probability =			

⋯ Formula			
Status	Not started		
Your score	0 / 1		0%
Response Let A and B be any two random independent events. Write down the formula for the joint probability of event A and event B. Note: • Correct answer: 1 point • Incorrect answer: 0 points • Unanswered: 0 points • Max score: 1 point • Min score: 0 points $P(A \cap B) =$			

 Conditional probability		
Status	Not started	
Your score	0 / 2	0%
Response In a University, 40% of the students play both football and volleyball, 60% of the students play football. What is the probability of a student playing volleyball given that he/she is already playing football? Please provide your answer with two digits after the decimal point. Note: • Correct answer: 2 points • Incorrect answer: 0 points • Unanswered: 0 points • Max score: 2 points • Min score: 0 points Probability =		

Knowledge Questions

☒ KQ: Question 1		
Status	Not started	
Your score	0 / 4	0%
Response Select all the techniques that can be used in visualizing high dimensional data. Note: • Correct answer: 1 point • Incorrect answer: -0.5 points • Unanswered: 0 points • Max score: 4 points • Min score: 0 points ☐ t-SNE ☐ UMAP ☐ PCA ☐ Latent Dirichlet Allocation ☐ Linear discriminant analysis ☐ SVM ☐ Random Forest		

☰ KQ: Question 2

Status Not started

Your score 0 / 12 0%

Response
Select 'Right' or 'Wrong' for the below statements.
Note:

- Correct answer: 2 points
- Incorrect answer: -1 points
- Unanswered: 0 points
- Max score: 12 points
- Min score: 0 points

Unanswered	Right	Wrong	
☐	☐	☐	PCA is an unsupervised data dimensionality reduction technique.
☐	☐	☐	In PCA, the principal axes are not always orthogonal.
☐	☐	☐	PCA works well only when the data is linearly co-related.
☐	☐	☐	PCA is not sensitive to scaling.
☐	☐	☐	No information is lost after PCA is applied on the data.
☐	☐	☐	PCA can be useful for reducing overfitting.

☰ KQ: Question 3

Status Not started

Your score 0 / 4 0%

Response
In sequential order, list down the steps that are performed in Principal Component Analysis.

0 word

☰ KQ: Question 4

Status Not started

Your score 0 / 5 0%

Response
When is data standardization required in Principal Component Analysis? Provide your answer in a single line.

0 word



Practice: Question 1

StatusNot started

Your score0 / 80%

Response

You are provided with two tables as shown below. Table 1 contains training data and Table 2 contains testing data. Your task is to Standardize the training and the testing data and fill the empty column with corresponding standardized values.

Please provide your answer with three digits after the decimal point. You can take three digits after the decimal point for standard deviation.

Note: Use Z-Score as a standardization technique.
Use the population variance for calculating the standard deviation.

- Correct answer: 1 point
- Incorrect answer: 0 points
- Unanswered: 0 points
- Max score: 8 points
- Min score: 0 points

Table 1

Raw Value	Standardized Value
56	
100	
23	
480	
32	

Table 2

Raw Value	Standardized Value
300	
10	
234	

Practice: Question 2

Status

Not started

Your score

0 / 12

0%

Response

Calculate the Covariance and Pearson correlation coefficient for the data given in the table below.

Note: Use sample Covariance formula for computing covariance.

Use sample standard deviation for calculating the Pearson correlation coefficient.

provide your answers with three digits after the decimal point. You can take three digits after the decimal point for standard deviation.

- Correct answer: 2 points
- Incorrect answer: 0 points
- Unanswered: 0 points
- Max score: 12 points
- Min score: 0 points

Height(H)	Weight(W)	Age(A)
5.4	52	21
5.6	58.5	26
4.8	42.6	23
5.11	60	29

Covariance(HW)= Covariance(WA)= Covariance(AH)= Corelation coefficient(HW)= Corelation coefficient(WA)= Corelation coefficient(AH)=

Practice: Question 3

Status

Not started

Your score

0 / 5

0%

Response

Compute the eigen values for the below matrix M .

Note:

- Correct answer: 5 points.
- Incorrect answer: 0 points
- Unanswered: 0 points
- Max score: 5 points
- Min score: 0 points

$$M = \begin{bmatrix} 5 & -3 \\ -6 & 2 \end{bmatrix}$$

Smaller Eigen value = Larger Eigen value =

 Programming

 Programming: Question 1

Status	Not started		
Your score	0 / 30		0%

Response

Please download the file iris.csv and perform the following tasks.

1. Create a user-defined function to compute pairwise Pearson correlation coefficients. The function takes two variables as arguments and returns the correlation coefficient between them. (6 points)
2. plot the graphs to visualize the correlation between two variables and also write a code to visualize the correlation matrix. (6 points).
3. Perform z-score standardization on the data. (6 points)
4. Compute covariance matrix for the standardized data. (6 points)
5. Find the eigenvalues and eigenvectors for the covariance matrix and print them in descending order. (6 points)

Note:

- Please upload your solution as one .ipynb file.
- If the code has any syntactical, logic errors, and/or exceptions it will not be accepted.

File

No file uploaded